POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Basic Electrical Engineering (New) | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

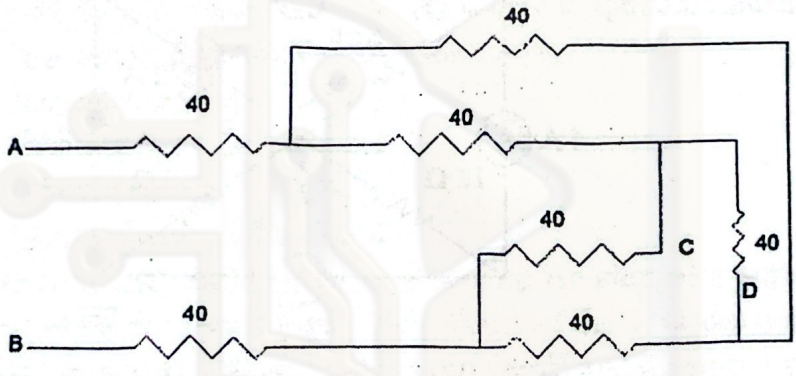
Attempt all the questions.

1. a) With the help of single line diagram, explain generation, transmission, and distribution of electricity in Nepal. 7

OR

Explain the present energy scenario in Nepal.

b) Why do we need color coding of Resistors? Find the equivalent resistance of the following circuit. 8



2. a) Determine the current in 5 Ω resistor for the circuit shown using Superposition Theorem (Figure 22). 8

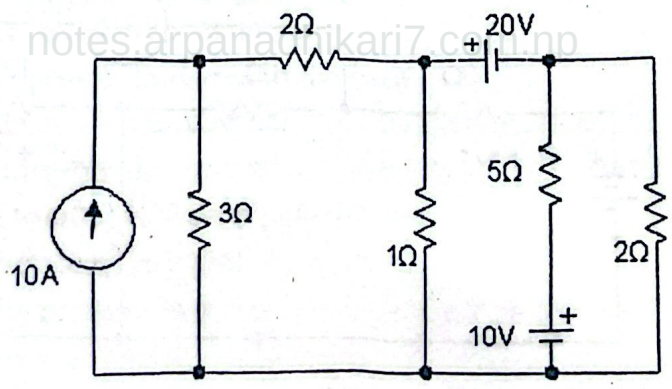
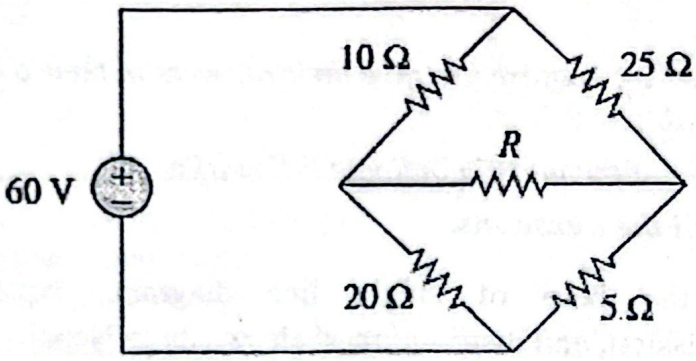


Figure 22

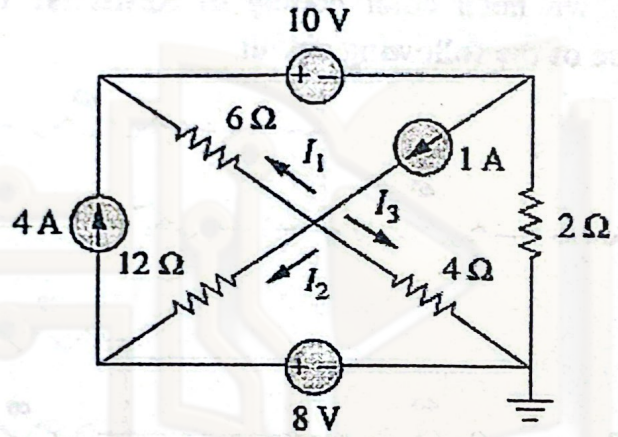
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b) Calculate the value of R that will absorb maximum power from the circuit (shown in the figure). Also calculate the maximum power drawn by it.



OR

Using mesh analysis, solve for I_1, I_2, I_3.



3. a) Determine the V_1 and V_2 using Nodal Analysis method. Also determine the current flowing through 30 Ω resistor (Figure 19) 7

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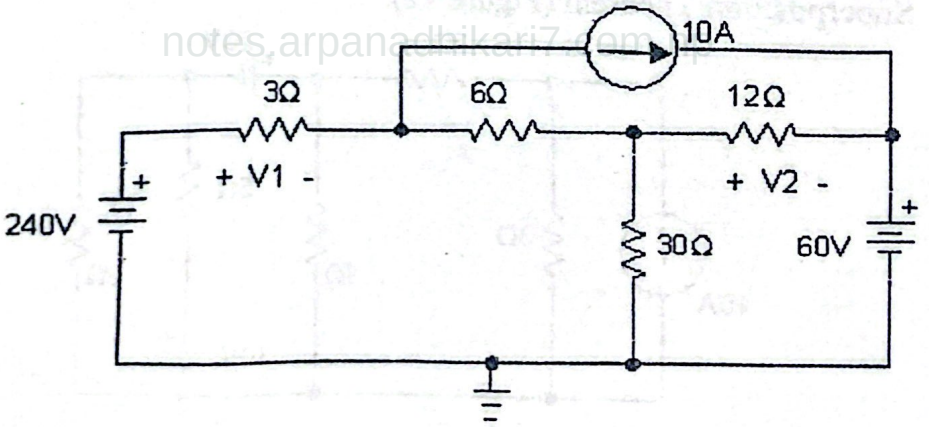


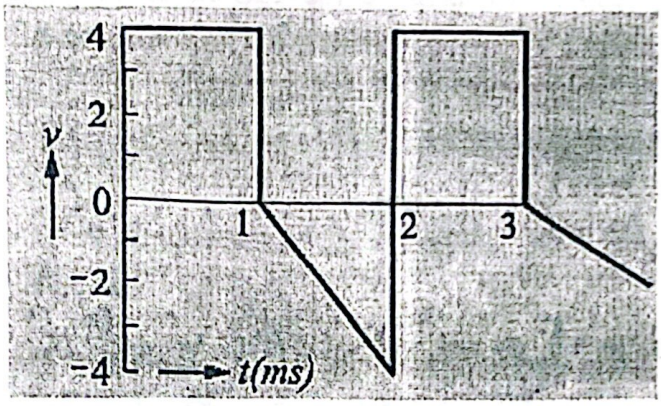
Figure 19

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power from the maximum power

b) Find the average value, rms value of the voltage waveform given below. 8



4. a) What will be the case for resonance in series RLC circuit? Derive the resonant frequency for this case. 7

b) A circuit have following electrical components in parallel: a) a resistor of 500 ohm, b) an inductance of 2H, c) capacitor of 10 microfarad. In this circuit 200V, 50 Hz source is applied. Calculate: 8

i. the total current drawn from the supply
ii. complex power
iii. active power
iv. reactive power
v. power factor of the circuit

5. a) The star connected stator of three phase, 50 Hz alternator supplies a balanced delta connected load. Each phase of the load consists of a coil of resistance of 15Ω and inductance 36mH and the phase voltage generated by the alternator is 231V. Calculate: 8

i. the phase and line currents
ii. the load power factor
iii. the power delivered to the load.

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b) An ideal 25 KVA transformer has 500 turns on the primary winding and 40 turns on the secondary winding. The primary winding is connected to a 3.3 KV supply. Calculate: 7

i. The secondary emf.
ii. The primary and secondary currents on full load.
iii. The maximum core flux.

6. a) Explain the operation of transformer on load and also draw phasor diagram at lagging power factor. 7

BEE-1st 2023 Fall

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b) A 220 Volts dc shunt motor draws a current of 5 A, and runs with a speed of 1500 rpm at no-load. When the motor is loaded, the speed drops to 1400 rpm. Calculate the current drawn by the motor at loaded condition. Given that the armature winding resistance is 0.2 Ω and field winding resistance is 110 Ω. 8

7. Write short notes on: **(Any two)** 2×5

a) Voltage divider Rule

b) Sources of Energy

c) Operation of DC motors

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